

Rahul Singh

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Profile

AI & Data Science undergraduate specializing in applied machine learning and data-driven product development. Experience in building end-to-end ML pipelines, performing EDA, feature engineering, model evaluation, interpretation, and deployment for real-world business and analytics problems.

Education

Dr. D. Y. Patil Institute of Engineering, Management & Research (DYPIEMR), Pune

B.E. in Artificial Intelligence & Data Science

2023 – 2027

CGPA: 8.55 / 10

Technical Skills

Languages: Python, C++, JavaScript

ML / Data: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Tools: TensorFlow, FastAPI, Streamlit, Git, GitHub, Jupyter Notebook

Core: Machine Learning, Data Science, EDA, Feature Engineering, Pipelines, Model Evaluation

Projects

Customer Churn Prediction System (End-to-End ML, Deployed)

2026

- Built end-to-end churn prediction system on Telco data (7k+ rows, 26 features) using pipelines and ColumnTransformer; reduced 6000+ features to 48.
- Merged multi-source datasets, removed data leakage features, fixed incorrect data types, and handled missing values using **Scikit-learn Pipelines** and **ColumnTransformer**.
- Trained Logistic Regression, Random Forest, Gradient Boosting; selected tuned Logistic Regression (class_weight=balanced) with **ROC-AUC $\approx$ 0.84**.
- Interpreted model to identify key churn drivers and deployed full pipeline as a **Streamlit web app** with probability and risk explanation.

IPL 50+ Score Prediction System (EDA $\rightarrow$ ML $\rightarrow$ Deployment)

2026

- Converted IPL batting dataset (15k+ rows) into ML problem; performed EDA, feature engineering, and data leakage prevention.
- Trained Logistic Regression, Random Forest, Gradient Boosting; final Gradient Boosting achieved **F1 $\approx$ 0.80, ROC-AUC $\approx$ 0.98**.
- Built pipeline-based training workflow and deployed using **Streamlit** for real-time predictions.

Drone Surveillance System for Anomaly Detection (AI Hackathon)

2025

- Built AI-based drone monitoring system for fire, flood, and anomaly detection using ML.
- Developed a simulation and real-time analytics pipeline using FastAPI and PyBullet.
- Qualified Round 2 (Top 120/240+) in Inno-Hackathon VIT 2025.

Achievements

- Qualified Round 2 (Top 120 out of 240+ teams) in Inno-Hackathon VIT 2025.
- Selected at internal college-level round of Smart India Hackathon (SIH) for AI-based drone surveillance project (VIRA).
- Built and deployed multiple end-to-end machine learning systems involving data preprocessing, feature engineering, model training, evaluation, and interpretation.